Quantitative Business Methods

What is decision modeling?

- Decision modeling is a scientific approach to decision making
- Various decision alternatives are available with different levels of risk and return
- Need to choose the right alternative from the set of alternatives available based on scientific decision tools

Types of decision models...

- Deterministic models: All relevant data values are known with certainty;
  - Eg. Assembling a computer
  - Technique/tool used: Linear programming

- Probabilistic models: Some data values are not known with certainty;
  - Eg. Sales of loafs of bread in a day
  - Technique/tool used: Probabilistic models

Objective of decision modeling...

- Decision making and improve its quality
- Identify optimal solution
- Integrating the systems
- Improve the objectivity of analysis
- Minimize the cost and maximize the profit
- Improve the productivity
- Success in competition and market leadership

Types of data: qualitative and quantitative...

Quantitative data are measures of values or counts and are expressed in numbers; Eg. Marks obtained in an examination

Qualitative data is concerned with description which can be observed but cannot be computed;
Eg. Importance of a subject

Steps involved in decision modeling...

Defining the problem

- Developing a model

Acquiring input data

- Developing a solution

Testing the solution

- Analysing the results and sensitivity analysis

Implementing the results

Formulation

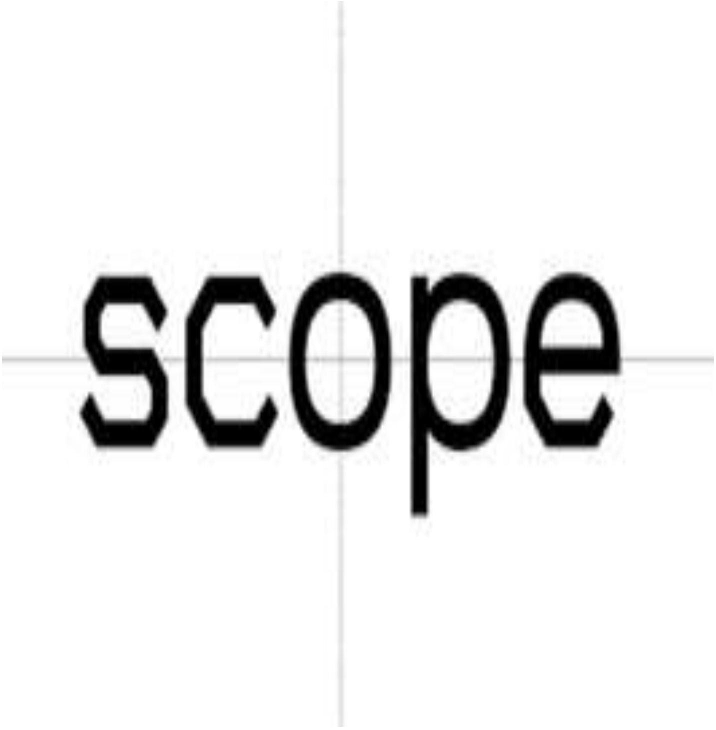
Solution

Interpretation

Advantages of a model...

- Problems under consideration become controllable through a model
- It provides a logical & systematic approach to the problem
- It provides the limitations and scope of the activity
- It helps in finding useful tools that eliminate duplication of methods applied to solve problems
- It helps in finding solutions for research and improvements in a system

Scope of decision modeling...



National plans and budget.

- Healthcare services and National Health Program.

Government development and public sector unit.

- Industrial establishment and private sector unit.

National defence services.

- Research & development.

Public works department.

- Business management.

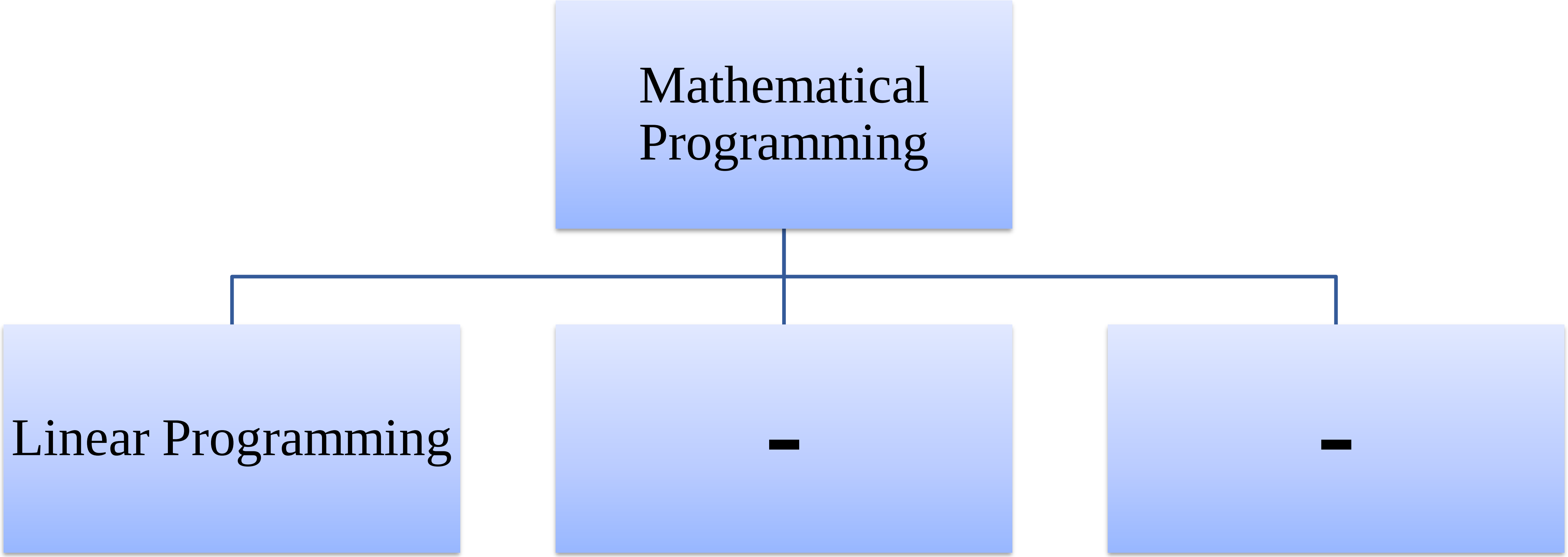
Agriculture and irrigation projects.

- Education & training.

Transport & Communication.

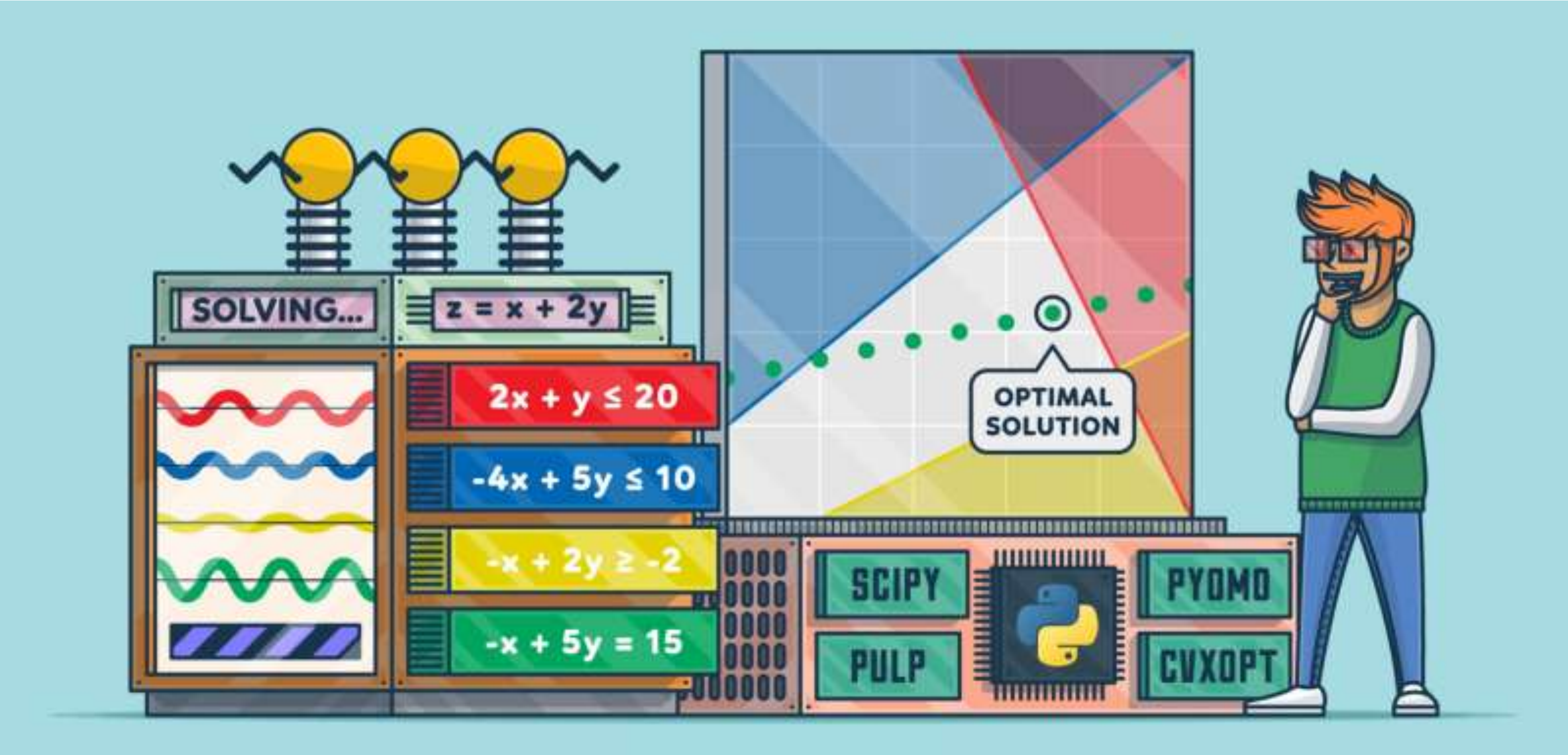
- Production management

Decision Modeling Technique



Linear Programming

- linear programming is a technique for the optimization of a linear objective function, subject to linear equality and linear inequality constraints.



Model Components

- **Decision variables** – mathematical symbols representing levels of activity of a firm
- **Objective function** - a linear mathematical relationship describing objective of a firm, in terms of decision variables - this function is to be maximized or minimized.
- **Constraints** – requirements or restrictions placed on the firm by the operating environment, stated in linear relationships of the decision variables
- **Parameters** - numerical coefficients and constants used in the objective function and constraints.

Formulation of linear programming model

- Objective function
- Constraints or restrictions
- Nonnegativity restriction

The problem formulation has the following steps:

Step1: Identifying the decision variables

Step2: Writing the objective function

Step3: Writing the constraints

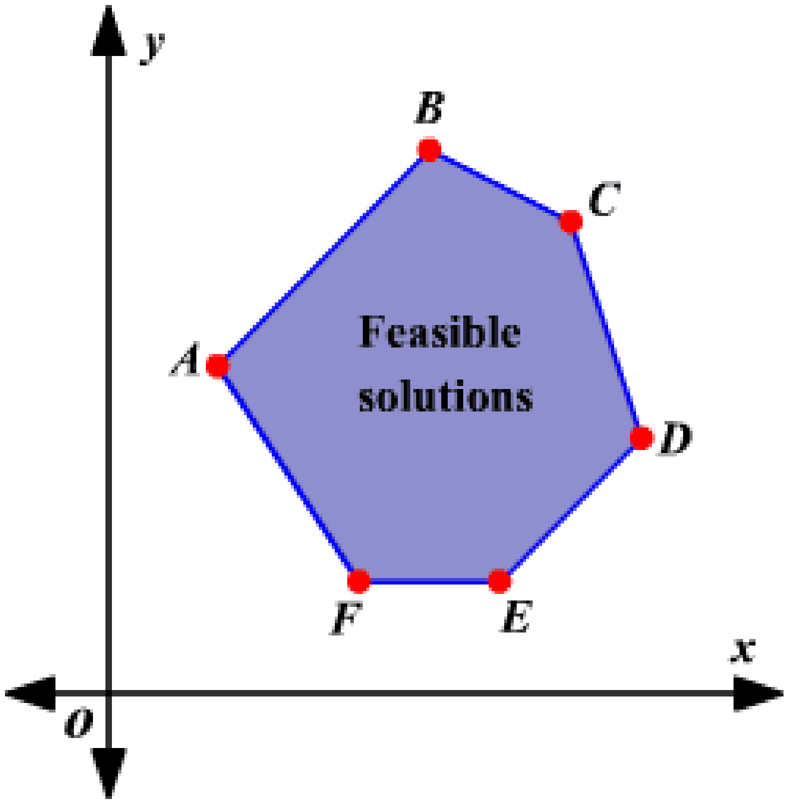
Step4: Writing the non-negativity restrictions

Steps of linear programming

Formulation: Expressing a problem scenario in terms of simple mathematical expressions

Solution: Solving mathematical expressions to find values for the variables

Interpretation: Understanding the solution obtained in order to evaluate the problem scenario



LP Problem Formulation: Example

Example:

Consider a small manufacturer making two products *A* and *B*.

Two resources R_1 and R_2 are required to make these products.

Each unit of Product *A* requires 1 unit of R_1 and 3 units of R_2. Each unit of Product *B* requires 1 unit of R_1 and 2 units of R_2.

The manufacturer has 5 units of R_1 and 12 units of R_2 available.

The manufacturer also makes a profit of Rs. 6 per unit of Product *A* sold and Rs. 5 per unit of Product *B* sold.

The manufacturer has to decide on the number of units of products A and B to produce such that profit has a maximum value.

LP Problem Formulation: Example

Identifying the decision variables

X = number of units of product A produced

Y = number of units of product B produced

Writing the objective function

Maximize $6X + 5Y$

Writing the constraints

$X + Y \leq 5$

$3X+2Y \leq 12$

Writing the non-negativity restrictions

$X \geq 0$

$Y \geq 0$

LP Problem Formulation: Example

Example:

A shop can make two types of sweets (A and B).

They use two resources-milk and sugar.

To make one packet of A, they need 3 kg of milk and 3 kg of sugar. To make one packet of B, they need 3 kg of milk and 4 kg of sugar.

They have 21 kg of milk and 28 kg of sugar.

These sweets are sold at Rs 1000 and 900 per packet respectively.

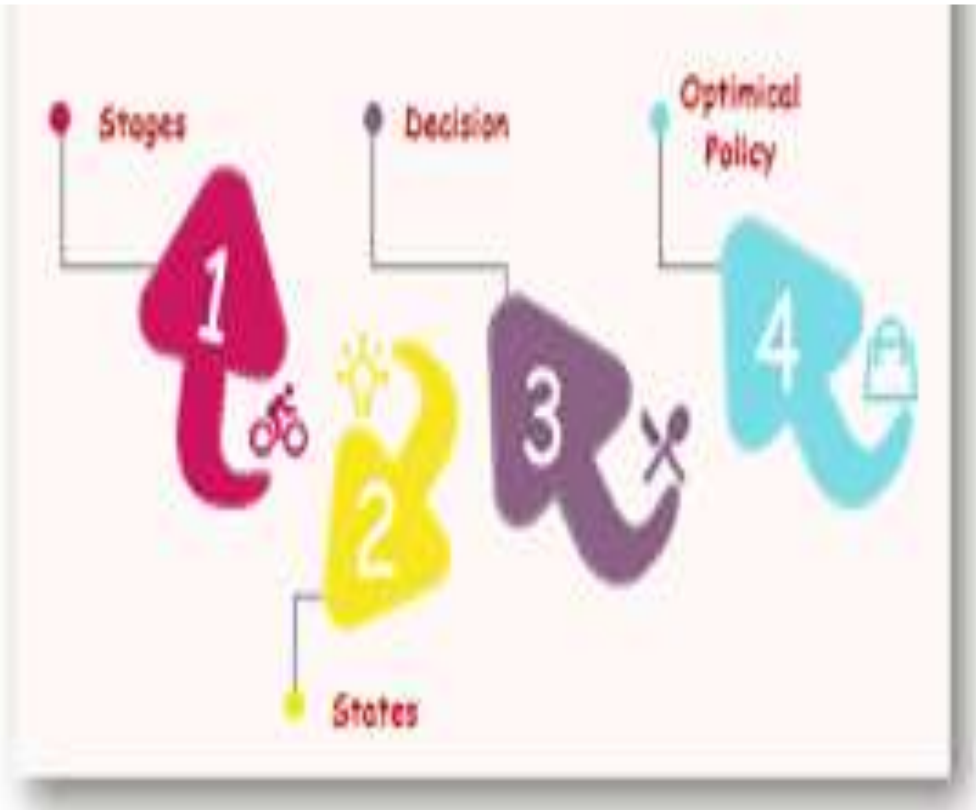
Find the product mix to maximize the revenue.

LP Problem Formulation: Example

Decision variables	
Objective function	
constraints	
Non-negativity restriction	

Properties of linear programming model

1. Problem seeks to maximize or minimize an objective
2. Constraints limit the degree to which the objective can be obtained
3. There must be alternative available
4. Mathematical relationships are linear (First degree equations)



Assumptions of linear programming model

- We assume that conditions of **certainty** exist.
- We also assume that **proportionality** exists in the objective function and constraints.
- The third assumption deals with **additivity.**
- We make the **divisibility** assumption.
- **Non-negative** variables

Merits and Demerits of Linear Programming

Merits

- **Optimal Resource Utilization**: LP ensures efficient allocation of limited resources (such as time, money, or materials) to achieve the desired outcome.
- **Simplicity and Clarity**: Linear programming models provide a clear mathematical representation of complex real-world problems
- **Wide Applicability**
- **Flexibility in Modeling**: LP accommodates diverse constraints and objectives, allowing businesses to model different scenarios effectively.
- **Improves Decision-Making**: LP provides quantitative solutions, helping decision-makers evaluate different options and choose the most efficient one.

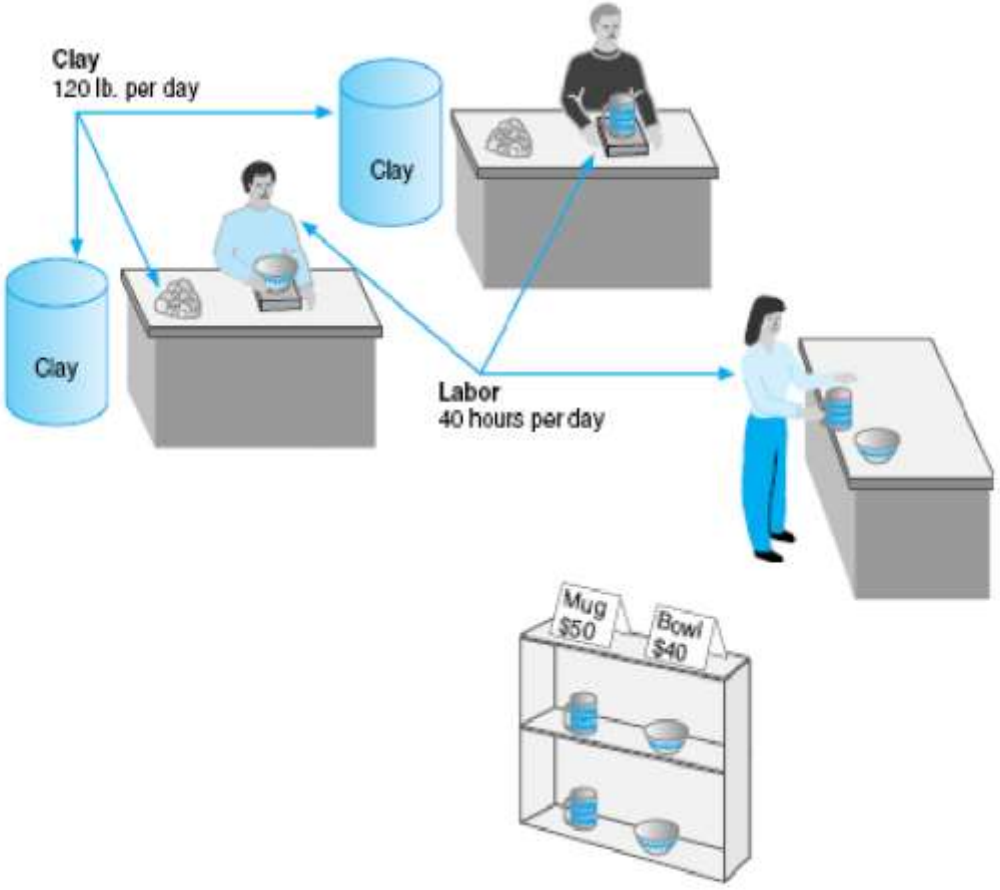
Merits and Demerits of Linear Programming

Demerits

- Linearity Assumption:
- Deterministic Nature:
- No Consideration for Qualitative Factors:
- Single Objective Optimization:
- Infeasibility and Unboundedness:

LP Problem Formulation: Example

Figure 2.1
Beaver Creek Pottery Company



LP problem Formulation : Example Contd……

- Product mix problem - Beaver Creek Pottery Company
- How many bowls and mugs should be produced to maximize profits given labor and materials constraints?
- Product resource requirements and unit profit:

	Resource Requirements		
Product	Labor (Hr./Unit)	Clay (Lb./Unit)	Profit ($/Unit)
Bowl	1	4	40
Mug	2	3	50

LP Problem Formulation : Example Contd……

Resource Availability:	40 hrs of labor per day 120 lbs of clay
Decision Variables:	x_1 = number of bowls to produce per day x_2 = number of mugs to produce per day
Objective Function:	Maximize $Z = \$40x_1 + \$50x_2$ Where Z = profit per day
Resource Constraints:	$1x_1 + 2x_2 \leq 40$ hours of labor $4x_1 + 3x_2 \leq 120$ pounds of clay
Non-Negativity Constraints:	$x_1 \geq 0$; $x_2 \geq 0$

LP Problem Formulation

Example. A retail store stocks two types of shirts *A* and *B*.
These are packed in attractive cardboard boxes.
During a week the store can sell a maximum of 400 shirts of type *A* and a maximum of 300 shirts of type *B*.
The storage capacity, however, is limited to a maximum of 600 of both types combined.
Type *A* shirt fetches a profit of Rs. 2/- per unit and type *B* a profit of Rs. 5/- per unit.
How many of each type the store should stock per week to maximize the total profit?
Formulate a mathematical model of the problem.

LP Problem Formulation

Example 6:
A patient consults a doctor to check up his ill health.
Doctor examines him and advises him that he is having deficiency of two vitamins, Vitamin A and Vitamin D.
Doctor advises him to consume vitamin A and D regularly for a period of time so that he can regain his health.
Doctor prescribes Tonic X and Tonic Y, which are having vitamin A, and D in certain proportion.
Also advises the patient to consume **at least** 40 units of Vitamin A and 50 units of Vitamin D daily.
The cost of tonics X and Y and the proportion of vitamin A and D that present in X and Y are given in the table below.
Formulate LPP to minimize the cost of tonics.

Vitamins	Tonic X	Tonic Y	Daily requirement in units
A	2	4	40
D	3	2	50
Cost in Rs per unit.	5	3	

LP Problem Formulation

Problem 2.4. A patient consult a doctor to check up his ill health. Doctor examines him and advises him that he is having deficiency of two vitamins, vitamin *A* and vitamin *D*. Doctor advises him to consume vitamin *A* and *D* regularly for a period of time so that he can regain his health. Doctor prescribes tonic *X* and tonic *Y*, which are having vitamin *A*, and *D* in certain proportion. Also advises the patient to consume **at least** 40 units of vitamin *A* and 50 units of vitamin Daily. The cost of tonics *X* and *Y* and the proportion of vitamin *A* and *D* that present in *X* and *Y* are given in the table below. Formulate l.p.p. to minimize the cost of tonics.

Vitamins	*Tonics*		*Daily requirement in units.*
	X	*Y*	
A	2	4	40
D	3	2	50
Cost in Rs. per unit.	5	3	

Decision Variable: Let amount of tonic Y used be x1, Let amount of tonic Y used be x2

Objective Function: Zmin = 5x1 + 3x2

Constraints: 2x1 + 4x2 $\geq$ 40 (Vitamin A); 3x1 + 2x2 $\geq$ 50 (Vitamin D)

Non-negativity restriction: x1, x2 $\geq$ 0

LP Problem Formulation

Example: A company manufactures two products, *X* and *Y* by using three machines *A*, *B*, and *C*.
Machine *A* has 4 hours of capacity available during the coming week.
Similarly, the available capacity of machines *B* and *C* during the coming week is 24 hours and 35 hours respectively.
One unit of product *X* requires one hour of Machine *A*, 3 hours of machine *B* and 10 hours of machine *C*.
Similarly, one unit of product *Y* requires 1 hour, 8 hour and 7 hours of machine *A*, *B* and *C* respectively.
When one unit of *X* is sold in the market, it yields a profit of Rs. 5/- per product and that of *Y* is Rs. 7/- per unit.
Formulate the problem.

LP Problem Formulation

Example: A company manufactures two products *X* and *Y*.
The profit contribution of *X* and *Y* are Rs.3/- and Rs. 4/- respectively.
The products *X* and *Y* require the services of four facilities.
The capacities of the four facilities *A*, *B*, *C*, and *D* are limited and the available capacities in hours are 200 Hrs, 150 Hrs, and 100 Hrs. and 80 hours respectively.
Product *X* requires 5, 3, 5 and 8 hours of facilities *A*, *B*, *C* and *D* respectively.
Similarly, the requirement of product *Y* is 4, 5, 5, and 4 hours respectively on *A*, *B*, *C* and *D*. Formulate the problem.

LP Problem Formulation

Example 7:

Reddy Mikks produces both interior and exterior paints from two raw materials, M1 and M2. The following tables provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	Exterior paint	Interior paint	
Raw material, M1	6	4	24
Raw material, M2	1	2	6
Profit per ton ($1000)	5	4	

A market survey indicates that the daily demand for interior paint cannot exceed that for the exterior paint by more than 1 ton.

Also, the maximum daily demand for interior paint is 2 tons.

Reddy Mikks wants to determine the optimum product mix of interior and exterior paints that maximizes the total daily profit.

LP Problem Formulation

Example 8:

Frontier Bakery have received order from company M/s Bodhraj Ltd. for the supply of high protein biscuits.

The order will require 1,000 kg of biscuits mix which is made from 4 ingredients R, S, T and U which cost Rs 16, Rs 4, Rs 6 and Rs 2 per kg respectively.

The batch must contain a minimum of 400 kg of protein, 250 kg of fat, 300 kg of carbohydrate and 50 kg of sugar. The ingredients contain the following percentage by weight:

Ingredients	Protein	Fat	Carbohydrate	Sugar	Filler
R	50%	30%	15%	5%	0%
S	10%	15%	50%	15%	10%
T	30%	5%	30%	30%	5%
U	0%	5%	5%	30%	60%

Only 150kg of 'S' and 200 Kg of 'T' are immediately available.

Draft a suitable LP Model.

LP Problem Formulation

Example 2·8. *Frontier Bakery have received order from a company M/s Bodhraj Ltd., for the supply of high protein biscuits. The order will require 1,000 kg. of biscuits mix which is made from 4 ingredients R, S, T and U which cost Rs. 16, Rs. 4, Rs. 6 and Rs. 2 per kg. respectively. The batch must contain a minimum of 400 kilos of protein, 250 kilos of fat, 300 kilos of carbohydrates and 50 kilos of sugar. The ingredients contain the following percentage by weight :*

Ingredients	Protein	Fat	Carbohydrate	Sugar	Filler
R	50%	30%	15%	5%	0%
S	10%	15%	50%	15%	10%
T	30%	5%	30%	30%	5%
U	0%	5%	5%	30%	60%

Only 150 kilos of S and 200 kilos of T are immediately available

Draft a suitable LP model.

Solution. By designating the number of units of ingredients *R*, *S*, *T* and *U* as *decision variables* x_1, x_2, x_3 and x_4 respectively, the appropriate mathematical formulation of the given problem can be expressed as *LP model* as shown below :

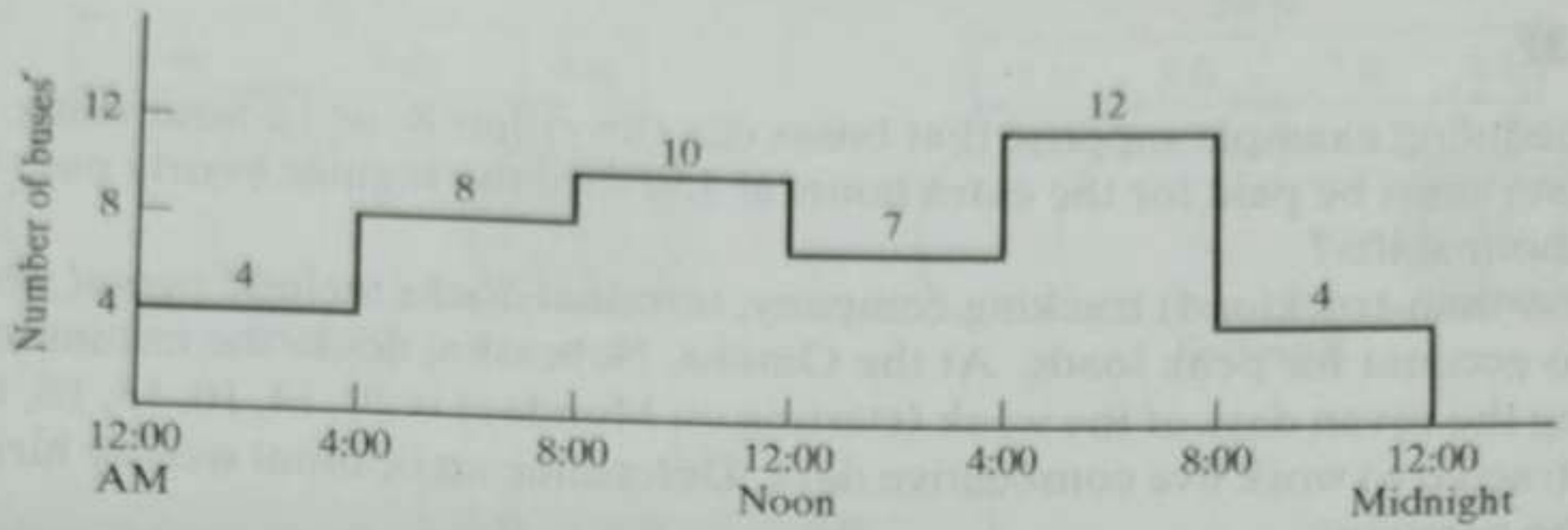
Minimize (total cost) $Z = 16x_1 + 4x_2 + 6x_3 + 2x_4$

Subject to the constraints

$$
\begin{aligned}
0.5\,x_1 + 0.1\,x_2 + 0.3x_3 + 0\,x_4 &\geq 400 \\
0.3\,x_1 + 0.15\,x_2 + 0.05x_3 + 0.05x_4 &\geq 250 \\
0.15\,x_1 + 0.5\,x_2 + 0.3\,x_3 + 0.05x_4 &\geq 300 \\
0.05\,x_1 + 0.15x_2 + 0.3\,x_3 + 0.3\,x_4 &\geq 50 \\
x_1 + x_2 + x_3 + x_4 &\geq 1{,}000 \\
x_2 &\leq 150 \\
x_3 &\leq 200 \\
x_1, x_2, x_3, x_4 &\geq 0
\end{aligned}
$$

Example 2.3-8 (Bus Scheduling)

Progress City is studying the feasibility of introducing a mass-transit bus system that will alleviate the smog problem by reducing in-city driving. The study seeks the minimum number of buses that can handle the transportation needs. After gathering necessary information, the city engineer noticed that the minimum number of buses needed fluctuated with the time of the day and that the required number of buses could be approximated by constant values over successive 4-hour intervals. Figure 2.8 summarizes the engineer's findings. To carry out the required daily maintenance, each bus can operate 8 successive hours a day only.



x_1 = number of buses starting at 12:01 A.M.
x_2 = number of buses starting at 4:01 A.M.
x_3 = number of buses starting at 8:01 A.M.
x_4 = number of buses starting at 12:01 P.M.
x_5 = number of buses starting at 4:01 P.M.
x_6 = number of buses starting at 8:01 P.M.

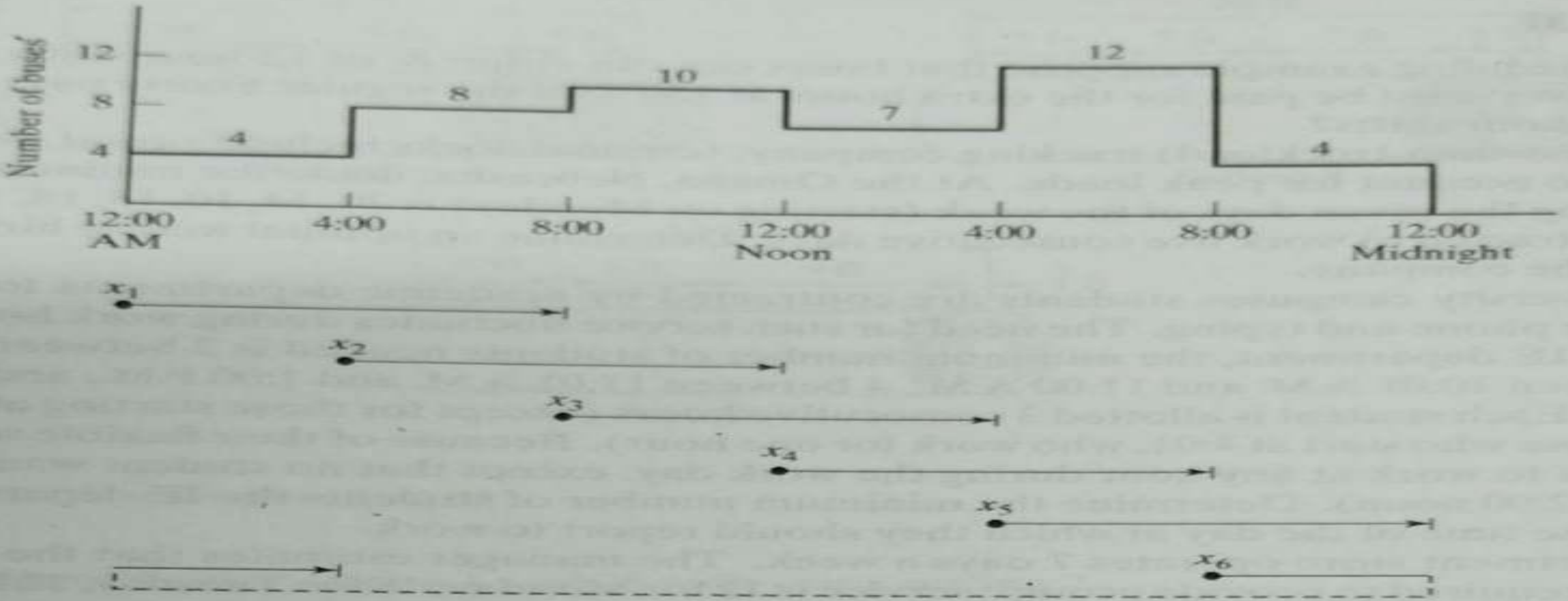


FIGURE 2.8
Number of buses as a function of the time of the day

Time period	Number of buses in operation
12:01 A.M. – 4:00 A.M.	$x_1 + x_6$
4:01 A.M. – 8:00 A.M.	$x_1 + x_2$
8:01 A.M. – 12:00 noon	$x_2 + x_3$
12:01 P.M. – 4:00 P.M.	$x_3 + x_4$
4:01 P.M. – 8:00 P.M.	$x_4 + x_5$
8:01 A.M. – 12:00 A.M.	$x_5 + x_6$

The complete model is thus written as

$$\text{Minimize } z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

subject to

$$\begin{aligned}
x_1 \qquad\qquad\qquad\qquad\quad + x_6 &\geq 4 \text{(12:01 A.M. – 4:00 A.M.)}\\
x_1 + x_2 \qquad\qquad\qquad\qquad &\geq 8 \text{(4:01 A.M. – 8:00 A.M.)}\\
x_2 + x_3 \qquad\qquad\qquad &\geq 10 \text{(8:01 A.M. – 12:0 noon)}\\
x_3 + x_4 \qquad\qquad &\geq 7 \text{(12:01 P.M. – 4:00 P.M.)}\\
x_4 + x_5 \qquad &\geq 12 \text{(4:01 P.M. – 8:00 P.M.)}\\
x_5 + x_6 &\geq 4 \text{(8:01 P.M. – 12:00 P.M.}\\
x_j \geq 0, j &= 1, 2, \ldots, 6
\end{aligned}$$

Solution: